

Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 17

File: results/dp_inf_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.716136363102116,0.06191074357045362,0.06,0.006000000000000001,0.004123646725745421,13.685996994018554,
3.5218226345062256
2,13.72660259513691,,,,,13.686008929443359,3.521824362945557
3,13.709877476344737,,,,,13.686023374938966,3.521826485824585
4,13.717380628722836,,,,,13.686035110473632,3.521828102493286
5,13.722983389297813,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686049906921387,
3.521830167007446
6,13.715259929181203,,,,,13.686062969970703,3.5218320011138915
7,13.711035469856222,,,,,13.686077563476562,3.5218339515686035
8,13.717059906595148,,,,,13.68608907470703,3.521835577774048
9,13.726437523744785,,,,,13.686100819396973,3.5218372089385985
10,13.716294993644347,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686113897705079
,3.5218390003204347
FINAL,,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686113897705079,3.521839000320
4347
```

File: results/dp_inf_alpha_0.5_dim_64_clients_100_seed_456.json

```
{
  "experiment_id": "dp_inf_alpha_0.5_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T02:11:17.955196",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.724946379484079,
      "test_metrics": {
        "mse": 13.688507598876953,
        "mae": 3.5223532188415527,
        "samples": 20000,
        "NDCG@10": 0.0577968700397596,
        "Hit@10": 0.09,
        "Precision@10": 0.009,
        "Recall@10": 0.0037602713721248985,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.726856706559197,
      "test_metrics": {
        "mse": 13.688492994689941,
        "mae": 3.5223507987976075,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ]
}
```

```
},
{
  "round": 3,
  "train_loss": 13.72317979983996,
  "test_metrics": {
    "mse": 13.688478712463379,
    "mae": 3.5223483833312987,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 13.717571462923006,
  "test_metrics": {
    "mse": 13.688463998413086,
    "mae": 3.5223459789276124,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 13.72191639376251,
  "test_metrics": {
    "mse": 13.688447169494628,
    "mae": 3.522343197631836,
    "samples": 20000,
    "NDCG@10": 0.0577968700397596,
    "Hit@10": 0.09,
    "Precision@10": 0.009,
    "Recall@10": 0.0037602713721248985,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.726362222109655,
  "test_metrics": {
    "mse": 13.688430990600587,
    "mae": 3.5223404735565187,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.720272413013747,
  "test_metrics": {
    "mse": 13.688412133789063,
    "mae": 3.522337431716919,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.720481877012725,
```

```

    "test_metrics": {
      "mse": 13.68839178314209,
      "mae": 3.5223342178344725,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 9,
    "train_loss": 13.719456072630846,
    "test_metrics": {
      "mse": 13.688371733093263,
      "mae": 3.5223310562133787,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 10,
    "train_loss": 13.720019055913703,
    "test_metrics": {
      "mse": 13.688349719238282,
      "mae": 3.5223275356292723,
      "samples": 20000,
      "NDCG@10": 0.057916521733220525,
      "Hit@10": 0.09,
      "Precision@10": 0.009,
      "Recall@10": 0.0037602713721248985,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
"final_metrics": {
  "mse": 13.688349719238282,
  "mae": 3.5223275356292723,
  "samples": 20000,
  "NDCG@10": 0.057916521733220525,
  "Hit@10": 0.09,
  "Precision@10": 0.009,
  "Recall@10": 0.0037602713721248985,
  "num_users_evaluated": 100
}
}

```

File: results/dp_inf_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.724946379484079,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688507598876953,3.5223532188415
527
2,13.726856706559197,,,,,13.688492994689941,3.5223507987976075
3,13.72317979983996,,,,,13.688478712463379,3.5223483833312987
4,13.717571462923006,,,,,13.688463998413086,3.5223459789276124
5,13.72191639376251,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688447169494628,3.52234319763183
6
6,13.726362222109655,,,,,13.688430990600587,3.5223404735565187
7,13.720272413013747,,,,,13.688412133789063,3.522337431716919
8,13.720481877012725,,,,,13.68839178314209,3.5223342178344725
9,13.719456072630846,,,,,13.688371733093263,3.5223310562133787
10,13.720019055913703,0.057916521733220525,0.09,0.009,0.0037602713721248985,13.688349719238282,3.5223275356
292723
FINAL,,0.057916521733220525,0.09,0.009,0.0037602713721248985,13.688349719238282,3.5223275356292723

```

File: results/dp_inf_alpha_1.0_dim_64_clients_100_seed_123.json

```
{
  "experiment_id": "dp_inf_alpha_1.0_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T03:03:53.002176",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 1.0,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.718662279494495,
      "test_metrics": {
        "mse": 13.685132504272461,
        "mae": 3.5215541271209716,
        "samples": 20000,
        "NDCG@10": 0.039105890351920955,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0023614718614718613,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.72615446777861,
      "test_metrics": {
        "mse": 13.6851140999121094,
        "mae": 3.521552025985718,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 13.720867875590036,
      "test_metrics": {
        "mse": 13.685094969177246,
        "mae": 3.5215497638702393,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 4,
      "train_loss": 13.730046955443349,
      "test_metrics": {
        "mse": 13.685073873901366,
        "mae": 3.521547235107422,
        "samples": 20000
      }
    }
  ],
}
```

```
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
},
{
  "round": 5,
  "train_loss": 13.717933644276929,
  "test_metrics": {
    "mse": 13.685054809570312,
    "mae": 3.5215449241638184,
    "samples": 20000,
    "NDCG@10": 0.039105890351920955,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.0023614718614718613,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.726125852344104,
  "test_metrics": {
    "mse": 13.685032702636718,
    "mae": 3.521542219543457,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.716970651184802,
  "test_metrics": {
    "mse": 13.685010177612305,
    "mae": 3.5215395736694335,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.714648433277425,
  "test_metrics": {
    "mse": 13.684988346862793,
    "mae": 3.52153685836792,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.718525544985368,
  "test_metrics": {
    "mse": 13.68496371307373,
    "mae": 3.5215338943481447,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
```

```

    },
    {
      "round": 10,
      "train_loss": 13.712659805782865,
      "test_metrics": {
        "mse": 13.684937284851074,
        "mae": 3.5215306037902834,
        "samples": 20000,
        "NDCG@10": 0.039105890351920955,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0023614718614718613,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ],
  "final_metrics": {
    "mse": 13.684937284851074,
    "mae": 3.5215306037902834,
    "samples": 20000,
    "NDCG@10": 0.039105890351920955,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.0023614718614718613,
    "num_users_evaluated": 100
  }
}

```

File: results/dp_inf_alpha_1.0_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.718662279494495,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.685132504272461,3.52155412712
09716
2,13.72615446777861,,,,,13.685114099121094,3.521552025985718
3,13.720867875590036,,,,,13.685094969177246,3.5215497638702393
4,13.730046955443349,,,,,13.685073873901366,3.521547235107422
5,13.717933644276929,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.685054809570312,3.52154492416
38184
6,13.726125852344104,,,,,13.685032702636718,3.521542219543457
7,13.716970651184802,,,,,13.685010177612305,3.5215395736694335
8,13.714648433277425,,,,,13.684988346862793,3.52153685836792
9,13.718525544985368,,,,,13.68496371307373,3.5215338943481447
10,13.712659805782865,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.684937284851074,3.5215306037
902834
FINAL,,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.684937284851074,3.5215306037902834

```

File: results/dp_inf_alpha_1.0_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_inf_alpha_1.0_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T03:02:12.285114",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 1.0,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,

```



```
"round": 6,
"train_loss": 13.73222987273084,
"test_metrics": {
  "mse": 13.686063023376464,
  "mae": 3.5218321056365967,
  "samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 7,
  "train_loss": 13.717393401576437,
  "test_metrics": {
    "mse": 13.686075192260741,
    "mae": 3.5218337844848633,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.7209430880483,
  "test_metrics": {
    "mse": 13.686087240600585,
    "mae": 3.5218354804992678,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.716816745731371,
  "test_metrics": {
    "mse": 13.686100651550293,
    "mae": 3.5218373363494875,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 13.719377080444453,
  "test_metrics": {
    "mse": 13.686112432861329,
    "mae": 3.521839047241211,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 13.686112432861329,
  "mae": 3.521839047241211,
  "samples": 20000,
```

```

    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  }
}

```

File: results/dp_inf_alpha_1.0_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.721815986879625,0.06191074357045362,0.06,0.006000000000000001,0.004123646725745421,13.685995581054687,
3.5218225173950195
2,13.73389563101359,,,,,13.686009576416016,3.5218245162963866
3,13.726155040955048,,,,,13.686022770690919,3.5218263854980467
4,13.717232466348856,,,,,13.686036561584473,3.521828374862671
5,13.717633917105777,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686050863647461,
3.521830379867554
6,13.7322987273084,,,,,13.686063023376464,3.5218321056365967
7,13.717393401576437,,,,,13.686075192260741,3.5218337844848633
8,13.7209430880483,,,,,13.686087240600585,3.5218354804992678
9,13.716816745731371,,,,,13.686100651550293,3.5218373363494875
10,13.719377080444453,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686112432861329
,3.521839047241211
FINAL,,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686112432861329,3.521839047241
211

```

File: results/dp_inf_alpha_1.0_dim_64_clients_100_seed_456.json

```

{
  "experiment_id": "dp_inf_alpha_1.0_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T03:05:31.166801",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 1.0,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.72330759866843,
      "test_metrics": {
        "mse": 13.688510116577149,
        "mae": 3.5223534229278566,
        "samples": 20000,
        "NDCG@10": 0.0577968700397596,
        "Hit@10": 0.09,
        "Precision@10": 0.009,
        "Recall@10": 0.0037602713721248985,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.719931132283033,
      "test_metrics": {
        "mse": 13.68849630279541,
        "mae": 3.522351094818115,
        "samples": 20000
      }
    }
  ]
}

```

```
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 13.726395302521766,
    "test_metrics": {
      "mse": 13.688480722045899,
      "mae": 3.5223484970092773,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 13.716910944022791,
    "test_metrics": {
      "mse": 13.688465667724609,
      "mae": 3.5223459297180177,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 13.724981207493943,
    "test_metrics": {
      "mse": 13.688449787902831,
      "mae": 3.522343238067627,
      "samples": 20000,
      "NDCG@10": 0.0577968700397596,
      "Hit@10": 0.09,
      "Precision@10": 0.009,
      "Recall@10": 0.0037602713721248985,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 6,
    "train_loss": 13.716727754199491,
    "test_metrics": {
      "mse": 13.688433050537109,
      "mae": 3.522340389251709,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 7,
    "train_loss": 13.717107462780824,
    "test_metrics": {
      "mse": 13.688415927124023,
      "mae": 3.5223376194000244,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
}
```

```

    }
  },
  {
    "round": 8,
    "train_loss": 13.721387370022974,
    "test_metrics": {
      "mse": 13.68839663848877,
      "mae": 3.522334504699707,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 9,
    "train_loss": 13.718756730316883,
    "test_metrics": {
      "mse": 13.688377203369141,
      "mae": 3.5223312973022463,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 10,
    "train_loss": 13.718138670994527,
    "test_metrics": {
      "mse": 13.68835595703125,
      "mae": 3.5223278732299805,
      "samples": 20000,
      "NDCG@10": 0.057916521733220525,
      "Hit@10": 0.09,
      "Precision@10": 0.009,
      "Recall@10": 0.0037602713721248985,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
"final_metrics": {
  "mse": 13.68835595703125,
  "mae": 3.5223278732299805,
  "samples": 20000,
  "NDCG@10": 0.057916521733220525,
  "Hit@10": 0.09,
  "Precision@10": 0.009,
  "Recall@10": 0.0037602713721248985,
  "num_users_evaluated": 100
}
}

```

File: results/dp_inf_alpha 1.0 dim 64 clients 100 seed 456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.72330759866843,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688510116577149,3.52235342292785
66
2,13.719931132283033,,,,,13.68849630279541,3.522351094818115
3,13.726395302521766,,,,,13.688480722045899,3.5223484970092773
4,13.716910944022791,,,,,13.688465667724609,3.5223459297180177
5,13.724981207493943,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688449787902831,3.5223432380676
27
6,13.716727754199491,,,,,13.688433050537109,3.522340389251709
7,13.717107462780824,,,,,13.688415927124023,3.5223376194000244
8,13.721387370022974,,,,,13.68839663848877,3.522334504699707
9,13.718756730316883,,,,,13.688377203369141,3.5223312973022463
10,13.718138670994527,0.057916521733220525,0.09,0.009,0.0037602713721248985,13.68835595703125,3.52232787322

```

99805

FINAL,,0.057916521733220525,0.09,0.009,0.0037602713721248985,13.68835595703125,3.5223278732299805

File: results_backup/attack_evaluation_summary.json

```
{
  "inf": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.54805,
      "accuracy": 0.545,
      "true_positive_rate": 0.48,
      "false_positive_rate": 0.39
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  },
  "8": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.53355,
      "accuracy": 0.515,
      "true_positive_rate": 0.6,
      "false_positive_rate": 0.57
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  },
  "4": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.5024500000000001,
      "accuracy": 0.505,
      "true_positive_rate": 0.61,
      "false_positive_rate": 0.6
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  },
  "2": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.5004,
      "accuracy": 0.5,
      "true_positive_rate": 0.64,
      "false_positive_rate": 0.64
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  },
  "1": {
    "mia": {
      "train_accuracy": 1.0,
      "auc": 0.48584999999999995,
      "accuracy": 0.51,
      "true_positive_rate": 0.62,
      "false_positive_rate": 0.6
    },
    "inversion": {
      "top_k_accuracy": 0.0,
      "top_k": 10,
      "num_users_evaluated": 50
    }
  }
}
```

```
}  
}  
}
```

File: results_backup/centralized_baseline.json

```
{  
  "experiment_id": "centralized_baseline",  
  "config": {  
    "num_users": 943,  
    "num_items": 1682,  
    "embedding_dim": 64,  
    "num_epochs": 50,  
    "learning_rate": 0.01,  
    "batch_size": 64,  
    "seed": 42,  
    "train_samples": 80000,  
    "test_samples": 20000  
  },  
  "train_losses": [  
    13.744499945831299,  
    13.733649127197266,  
    13.7224368309021,  
    13.710547796630859,  
    13.697592328643799,  
    13.683081722259521,  
    13.666377948760987,  
    13.646623229217528,  
    13.622682015228271,  
    13.593017753601075,  
    13.555565504455567,  
    13.507558028411864,  
    13.445319319915772,  
    13.364007292175293,  
    13.257323151397705,  
    13.117280011749267,  
    12.93408091430664,  
    12.696154238128662,  
    12.390833408355713,  
    12.005739274597168,  
    11.53115025177002,  
    10.963267489624023,  
    10.307968677902222,  
    9.583248323822021,  
    8.81859575920105,  
    8.050384371566773,  
    7.314124571609497,  
    6.636849789810181,  
    6.033216427993774,  
    5.506068143081665,  
    5.050083262443542,  
    4.655981608772278,  
    4.3136439014434815,  
    4.0142961641311645,  
    3.7506541988372804,  
    3.5169398542404173,  
    3.3086113360404967,  
    3.1220811665534973,  
    2.954416963863373,  
    2.803191617488861,  
    2.6663891534805297,  
    2.5422803891181944,  
    2.4293844024658204,  
    2.3264532668113707,  
    2.2323575150489807,  
    2.146175816440582,  
    2.0670647891044616,  
    1.9942732004165649,  
    1.9271537218093873,  
    1.865168280506134  
  ],  
  "test_metrics_history": [  
    {  
      "epoch": 10,  

```

```
"metrics": {
  "mse": 13.622893885803222,
  "mae": 3.5133981464385986,
  "accuracy": 0.0,
  "samples": 20000,
  "NDCG@10": 0.06899745675111287,
  "Hit@10": 0.1,
  "Precision@10": 0.011000000000000001,
  "Recall@10": 0.00757346677004221,
  "num_users_evaluated": 100
},
{
  "epoch": 20,
  "metrics": {
    "mse": 11.968002226257324,
    "mae": 3.2785331844329835,
    "accuracy": 0.0029,
    "samples": 20000,
    "NDCG@10": 0.24605057094704916,
    "Hit@10": 0.43,
    "Precision@10": 0.057999999999999996,
    "Recall@10": 0.027724766988414317,
    "num_users_evaluated": 100
  }
},
{
  "epoch": 30,
  "metrics": {
    "mse": 5.515742160415649,
    "mae": 2.0133992692947387,
    "accuracy": 0.0357,
    "samples": 20000,
    "NDCG@10": 0.28360105825565596,
    "Hit@10": 0.46,
    "Precision@10": 0.071000000000000001,
    "Recall@10": 0.035696776316918345,
    "num_users_evaluated": 100
  }
},
{
  "epoch": 40,
  "metrics": {
    "mse": 2.956100328826904,
    "mae": 1.3657585359573363,
    "accuracy": 0.04945,
    "samples": 20000,
    "NDCG@10": 0.2798283914482264,
    "Hit@10": 0.49,
    "Precision@10": 0.072000000000000001,
    "Recall@10": 0.04163180844534197,
    "num_users_evaluated": 100
  }
},
{
  "epoch": 50,
  "metrics": {
    "mse": 2.0449277555465697,
    "mae": 1.1063344123840333,
    "accuracy": 0.05495,
    "samples": 20000,
    "NDCG@10": 0.22501065427109246,
    "Hit@10": 0.38,
    "Precision@10": 0.056000000000000001,
    "Recall@10": 0.03393271862353286,
    "num_users_evaluated": 100
  }
}
],
"final_metrics": {
  "mse": 2.0449277555465697,
  "mae": 1.1063344123840333,
  "accuracy": 0.05495,
  "samples": 20000,
```

```
"NDCG@10": 0.22501065427109246,  
"Hit@10": 0.38,  
"Precision@10": 0.056000000000000001,  
"Recall@10": 0.03393271862353286,  
"num_users_evaluated": 100  
}  
}
```

File: results_backup/dp_1_alpha_0.5_dim_64_clients_100_seed_123.json

```
{  
  "experiment_id": "dp_1_alpha_0.5_dim_64_clients_100_seed_123",  
  "timestamp": "2026-02-11T02:52:32.667743",  
  "config": {  
    "num_users": 943,  
    "num_items": 1682,  
    "embedding_dim": 64,  
    "num_clients": 100,  
    "alpha": 0.5,  
    "dp_epsilon": 1,  
    "use_dp": true,  
    "dp_sigma": 75.0625,  
    "dp_clip_norm": 1.0,  
    "num_rounds": 10,  
    "local_epochs": 3,  
    "learning_rate": 0.01,  
    "batch_size": 32,  
    "seed": 123  
  },  
  "rounds": [  
    {  
      "round": 1,  
      "train_loss": 38963.97423880543,  
      "test_metrics": {  
        "mse": 26.465855584716795,  
        "mae": 4.226308657836914,  
        "samples": 20000,  
        "NDCG@10": 0.04521064059975586,  
        "Hit@10": 0.07,  
        "Precision@10": 0.006999999999999999,  
        "Recall@10": 0.0017224435186132148,  
        "num_users_evaluated": 100  
      },  
      "aggregation": {  
        "num_clients": 100,  
        "total_samples": 80000  
      }  
    },  
    {  
      "round": 2,  
      "train_loss": 40377.15169415534,  
      "test_metrics": {  
        "mse": 63.29663121948242,  
        "mae": 6.364564102935791,  
        "samples": 20000  
      },  
      "aggregation": {  
        "num_clients": 100,  
        "total_samples": 80000  
      }  
    },  
    {  
      "round": 3,  
      "train_loss": 41567.93433565475,  
      "test_metrics": {  
        "mse": 125.221526171875,  
        "mae": 8.902955809020996,  
        "samples": 20000  
      },  
      "aggregation": {  
        "num_clients": 100,  
        "total_samples": 80000  
      }  
    }  
  ],  
}
```

```
{
  "round": 4,
  "train_loss": 43131.80167295857,
  "test_metrics": {
    "mse": 213.16070805664063,
    "mae": 11.645036456298827,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 44382.366078070045,
  "test_metrics": {
    "mse": 327.51094931640625,
    "mae": 14.392618478393555,
    "samples": 20000,
    "NDCG@10": 0.04050924333460803,
    "Hit@10": 0.03,
    "Precision@10": 0.0030000000000000005,
    "Recall@10": 0.002470653377630122,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 45507.749855667615,
  "test_metrics": {
    "mse": 459.7774859375,
    "mae": 17.018246719360352,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 46708.978485058025,
  "test_metrics": {
    "mse": 622.90454921875,
    "mae": 19.758551049804687,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 47813.48624437197,
  "test_metrics": {
    "mse": 812.9470924804688,
    "mae": 22.52230757446289,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 49770.981168050545,
  "test_metrics": {
```

```

        "mse": 1017.7248899414062,
        "mae": 25.38714585571289,
        "samples": 20000
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
},
{
    "round": 10,
    "train_loss": 50824.836797064934,
    "test_metrics": {
        "mse": 1264.3574362304687,
        "mae": 28.299958532714843,
        "samples": 20000,
        "NDCG@10": 0.038747436890763806,
        "Hit@10": 0.05,
        "Precision@10": 0.005,
        "Recall@10": 0.003423709984180309,
        "num_users_evaluated": 100
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
}
},
"final_metrics": {
    "mse": 1264.3574362304687,
    "mae": 28.299958532714843,
    "samples": 20000,
    "NDCG@10": 0.038747436890763806,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.003423709984180309,
    "num_users_evaluated": 100
}
}

```

File: results_backup/dp_1_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,38963.97423880543,0.04521064059975586,0.07,0.006999999999999999,0.0017224435186132148,26.465855584716795,
4.226308657836914
2,40377.15169415534,,,,,63.29663121948242,6.364564102935791
3,41567.93433565475,,,,,125.221526171875,8.902955809020996
4,43131.80167295857,,,,,213.16070805664063,11.645036456298827
5,44382.366078070045,0.04050924333460803,0.03,0.0030000000000000005,0.002470653377630122,327.51094931640625
,14.392618478393555
6,45507.749855667615,,,,,459.7774859375,17.018246719360352
7,46708.978485058025,,,,,622.90454921875,19.758551049804687
8,47813.48624437197,,,,,812.9470924804688,22.52230757446289
9,49770.981168050545,,,,,1017.7248899414062,25.38714585571289
10,50824.836797064934,0.038747436890763806,0.05,0.005,0.003423709984180309,1264.3574362304687,28.2999585327
14843
FINAL,,0.038747436890763806,0.05,0.005,0.003423709984180309,1264.3574362304687,28.299958532714843

```

File: results_backup/dp_1_alpha_0.5_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_1_alpha_0.5_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:49:28.884828",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 1,
    "use_dp": true,
    "dp_sigma": 75.0625,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,

```

```
"local_epochs": 3,
"learning_rate": 0.01,
"batch_size": 32,
"seed": 42
},
"rounds": [
  {
    "round": 1,
    "train_loss": 38960.01599917924,
    "test_metrics": {
      "mse": 26.324150482177735,
      "mae": 4.210136761856079,
      "samples": 20000,
      "NDCG@10": 0.03961790233246403,
      "Hit@10": 0.07,
      "Precision@10": 0.008,
      "Recall@10": 0.003440055417135213,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 2,
    "train_loss": 40141.44112942695,
    "test_metrics": {
      "mse": 62.564993969726565,
      "mae": 6.311017587280274,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 41667.38061528948,
    "test_metrics": {
      "mse": 124.27513557128906,
      "mae": 8.879233229064942,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 42687.622308907325,
    "test_metrics": {
      "mse": 210.34479968261718,
      "mae": 11.537402191162109,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 44143.33736494178,
    "test_metrics": {
      "mse": 317.18783852539065,
      "mae": 14.235216018676757,
      "samples": 20000,
      "NDCG@10": 0.04961254792901989,
      "Hit@10": 0.08,
      "Precision@10": 0.009000000000000001,
      "Recall@10": 0.005139105992047169,
```

```
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 45012.75422023619,
  "test_metrics": {
    "mse": 449.4102943359375,
    "mae": 16.89489171142578,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 47048.08474836159,
  "test_metrics": {
    "mse": 616.3688580078125,
    "mae": 19.67854136657715,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 48080.34391334988,
  "test_metrics": {
    "mse": 778.6992239257812,
    "mae": 22.125241497802733,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 49151.029420506144,
  "test_metrics": {
    "mse": 983.3803299804688,
    "mae": 24.801917776489258,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 50547.670809926174,
  "test_metrics": {
    "mse": 1219.79353671875,
    "mae": 27.73233918762207,
    "samples": 20000,
    "NDCG@10": 0.06373179327591641,
    "Hit@10": 0.06,
    "Precision@10": 0.007000000000000001,
    "Recall@10": 0.0022059073478178253,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
```

```

    "total_samples": 80000
  }
},
"final_metrics": {
  "mse": 1219.79353671875,
  "mae": 27.73233918762207,
  "samples": 20000,
  "NDCG@10": 0.06373179327591641,
  "Hit@10": 0.06,
  "Precision@10": 0.007000000000000001,
  "Recall@10": 0.0022059073478178253,
  "num_users_evaluated": 100
},
"attack_results": {
  "mia": {
    "train_accuracy": 1.0,
    "auc": 0.48584999999999995,
    "accuracy": 0.51,
    "true_positive_rate": 0.62,
    "false_positive_rate": 0.6
  },
  "inversion": {
    "top_k_accuracy": 0.0,
    "top_k": 10,
    "num_users_evaluated": 50
  }
}
}

```

File: results_backup/dp_1_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,38960.01599917924,0.03961790233246403,0.07,0.008,0.003440055417135213,26.324150482177735,4.21013676185607
2,40141.44112942695,,,,,62.564993969726565,6.311017587280274
3,41667.38061528948,,,,,124.27513557128906,8.879233229064942
4,42687.622308907325,,,,,210.34479968261718,11.537402191162109
5,44143.33736494178,0.04961254792901989,0.08,0.009000000000000001,0.005139105992047169,317.18783852539065,1
4.235216018676757
6,45012.75422023619,,,,,449.4102943359375,16.89489171142578
7,47048.08474836159,,,,,616.3688580078125,19.67854136657715
8,48080.34391334988,,,,,778.6992239257812,22.125241497802733
9,49151.029420506144,,,,,983.3803299804688,24.801917776489258
10,50547.670809926174,0.06373179327591641,0.06,0.007000000000000001,0.0022059073478178253,1219.79353671875,
27.73233918762207
FINAL,,0.06373179327591641,0.06,0.007000000000000001,0.0022059073478178253,1219.79353671875,27.732339187622
07

```

File: results_backup/dp_1_alpha_0.5_dim_64_clients_100_seed_456.json

```

{
  "experiment_id": "dp_1_alpha_0.5_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T02:55:35.345975",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 1,
    "use_dp": true,
    "dp_sigma": 75.0625,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 38904.826062472566,

```

```
"test_metrics": {
  "mse": 25.98537597351074,
  "mae": 4.186658390045166,
  "samples": 20000,
  "NDCG@10": 0.040225689933686935,
  "Hit@10": 0.07,
  "Precision@10": 0.008,
  "Recall@10": 0.0035781231096256656,
  "num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 2,
  "train_loss": 40290.495299244,
  "test_metrics": {
    "mse": 62.07684714355469,
    "mae": 6.308922241210937,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 41192.94593482481,
  "test_metrics": {
    "mse": 120.73841142578125,
    "mae": 8.752308038330078,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 42551.16011562006,
  "test_metrics": {
    "mse": 206.5189384765625,
    "mae": 11.415081701660156,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 43977.35872124249,
  "test_metrics": {
    "mse": 312.61522465820315,
    "mae": 14.057603115844726,
    "samples": 20000,
    "NDCG@10": 0.04299470147220183,
    "Hit@10": 0.06,
    "Precision@10": 0.0070000000000000001,
    "Recall@10": 0.005984292189093961,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
```

```
"train_loss": 45112.42385186509,
"test_metrics": {
  "mse": 446.0535176757812,
  "mae": 16.764670126342775,
  "samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 7,
  "train_loss": 46971.01330695465,
  "test_metrics": {
    "mse": 600.2338583007812,
    "mae": 19.36639291381836,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 47725.49999426106,
  "test_metrics": {
    "mse": 780.3972463867187,
    "mae": 22.106222232055664,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 49063.3764904598,
  "test_metrics": {
    "mse": 976.159595703125,
    "mae": 24.826724716186522,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 50430.77642351835,
  "test_metrics": {
    "mse": 1200.692373046875,
    "mae": 27.474232513427733,
    "samples": 20000,
    "NDCG@10": 0.03762802832552227,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.002745130885374788,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 1200.692373046875,
  "mae": 27.474232513427733,
  "samples": 20000,
  "NDCG@10": 0.03762802832552227,
```

```
"Hit@10": 0.05,  
"Precision@10": 0.005,  
"Recall@10": 0.002745130885374788,  
"num_users_evaluated": 100  
}  
}
```

File: results_backup/dp_1_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

```
-----  
Round, Train_Loss, NDCG@10, Hit@10, Precision@10, Recall@10, MSE, MAE  
1, 38904.826062472566, 0.040225689933686935, 0.07, 0.008, 0.0035781231096256656, 25.98537597351074, 4.186658390045  
166  
2, 40290.495299244, , , , , 62.07684714355469, 6.308922241210937  
3, 41192.94593482481, , , , , 120.73841142578125, 8.752308038330078  
4, 42551.16011562006, , , , , 206.5189384765625, 11.415081701660156  
5, 43977.35872124249, 0.04299470147220183, 0.06, 0.007000000000000001, 0.005984292189093961, 312.61522465820315, 1  
4.057603115844726  
6, 45112.42385186509, , , , , 446.0535176757812, 16.764670126342775  
7, 46971.01330695465, , , , , 600.2338583007812, 19.36639291381836  
8, 47725.49999426106, , , , , 780.3972463867187, 22.106222232055664  
9, 49063.3764904598, , , , , 976.159595703125, 24.826724716186522  
10, 50430.77642351835, 0.03762802832552227, 0.05, 0.005, 0.002745130885374788, 1200.692373046875, 27.4742325134277  
33  
FINAL, , 0.03762802832552227, 0.05, 0.005, 0.002745130885374788, 1200.692373046875, 27.474232513427733
```